

Hypothesis Testing

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Previously...

- lect02 - fundamentals of statistics
- lect03 - a biological question $\rightarrow$ a computational one

Today: learning objective

- 1 Identify hypothesis testing problems
- 2 Understand what hypothesis testing is
- 3 Apply hypothesis testing procedures

Today's lecture

- 1 Motivation and Problem definition
- 2 Key concepts
- 3 An example of hypothesis testing
- 4 Discussion

Obs #1: breast cancer subtypes by hormone receptors

Estrogen receptor (ER) protein expression affects breast cancer prognosis (EBCTCG et al. 2011; Cancer Genome Atlas Network 2012).

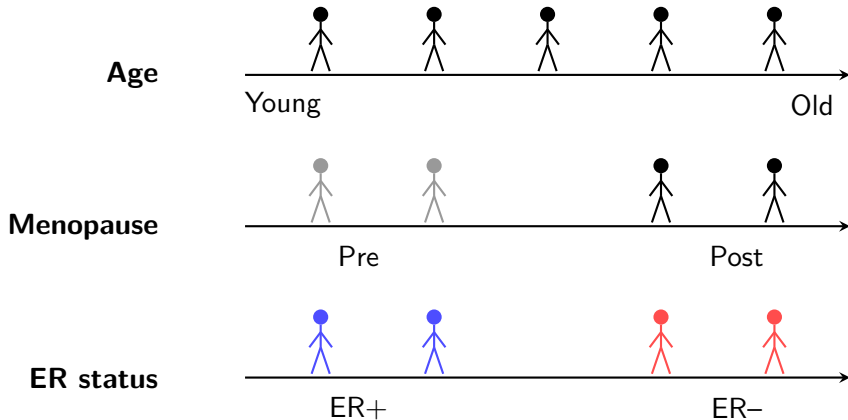
- **ER+** (estrogen receptor positive): ~70% of breast cancers
 - Treatment: hormone therapy (e.g., tamoxifen)
 - Better prognosis
- **ER-** (Estrogen Receptor negative): ~30% of breast cancers
 - Treatment: chemotherapy
 - More aggressive, poor outcomes

Obs #2: age and menopause effects on estrogen

Menopause (~50 years) dramatically changes estrogen levels

- Pre-menopausal: high circulating estrogen
- Post-menopausal: low circulating estrogen
- Age-correlated gene expression in breast cancer is largely regulated by **endocrine (estrogen) signaling** (Osako et al. 2020)

Three axes of variation



Key question: multiple sources of variation

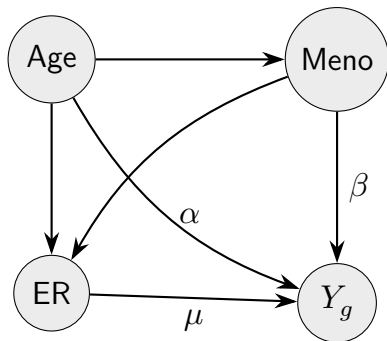
- Can we identify multiple independent disease mechanisms?
- How do they manifest in observed transcriptional profiles?

- ① **ER status:** Tumor-intrinsic differences (ER+ vs ER-)
- ② **Age:** age effects in general population
- ③ **Menopause:** pre- vs. post-menopause

They are all correlated... but not necessarily the same

- Post-menopausal but young (e.g., surgical menopause)
- Pre-menopausal but older (late menopause)

Probabilistic model



Observed variables:

- Age (continuous), Menopause (categorical), ER status (categorical)
- Gene expression matrix Y_{gi} for a gene g in a sample i .

Parameters (open):

- μ : ER status effect
- α : direct age effect
- β : menopause effect

E.g., TCGA BRCA data downloaded via recount3

```
library(recount3) ## install via Bioconductor

if.needed(.file, {
  ## Take TCGA
  tcga.proj <- subset(available_projects(),
                     file_source == "tcga")
  ## Choose BRCA (breast cancer)
  brca.info <- tcga.proj[tcga.proj$project == "BRCA", ]
  ## Save the object
  rse.brca <- create_rse(brca.info)
  saveRDS(rse.brca, .file)
})

## We can quickly read in `.rds` in the memory
rse <- readRDS(.file)
```

Side: a trick to avoid overwriting

```
if.needed <-  
  function(.file, .code) {  
    if(!all(file.exists(unlist(.file)))){ .code }  
    stopifnot(all(file.exists(unlist(.file))))  
  }
```

Side: What is RangedSummarizedExperiment object?

		samples (columns)				
		S1	S2	S3	...	
genes (rows)	G1	10	25	18	...	← assays (counts, TPM)
	G2	5	12	8	...	
	...	...	...	...	...	

rowRanges: genomic coordinates (chr, start, end, strand)

colData: sample metadata (tumor/normal, clinical info)

Column (sample) data

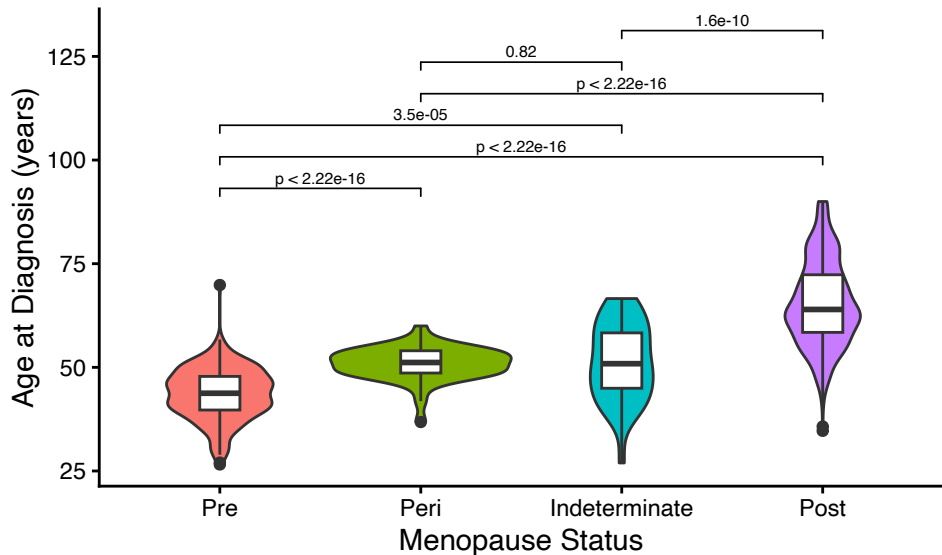
```
clinical <-  
  colData(rse) %>%  
  as.data.frame() %>%  
  select(  
    sample.id = tcga.tcga_barcode,  
    sample.type = tcga.cgc_sample_sample_type,  
    age = tcga.gdc_cases.diagnoses.age_at_diagnosis,  
    menopause = tcga.xml_menopause_status,  
    tumor.stage = tcga.gdc_cases.diagnoses.tumor_stage,  
    ER.status = tcga.xml_breast_carcinoma_estrogen_receptor_status,  
    PR.status = tcga.xml_breast_carcinoma_progesterone_receptor_status,  
    HER2.status = tcga.xml_lab_proc_her2_neu_immunohistochemistry_receptor  
  ) %>%  
  as.data.table()
```

Header lines

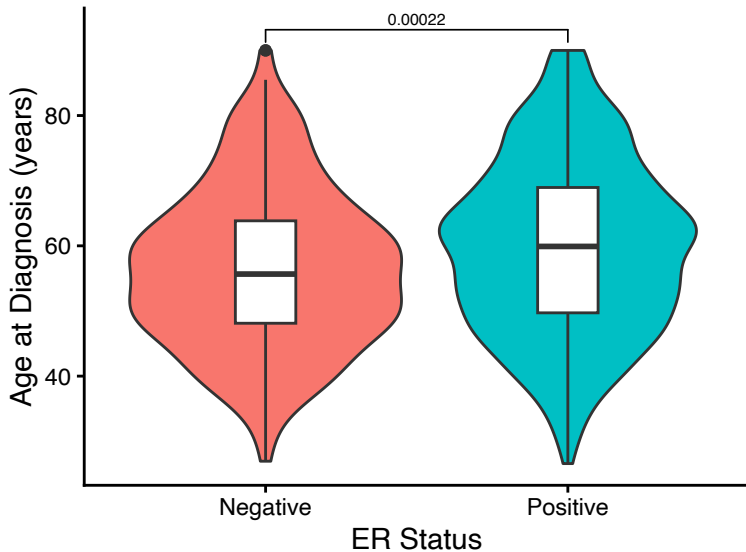
```
head(clinical, 5)
```

```
##           sample.id  sample.type  age
##           <char>      <char> <int>
## 1: TCGA-E9-A249-01A-11R-A169-07 Primary Tumor 16689
## 2: TCGA-BH-A1EX-01A-11R-A13Q-07 Primary Tumor 24544
## 3: TCGA-AO-A12A-01A-21R-A115-07 Primary Tumor 17420
## 4: TCGA-AC-A3OD-01B-06R-A220-07 Primary Tumor 25186
## 5: TCGA-AC-A23G-01A-11R-A213-07 Primary Tumor 27934
##
##                                     menopause
##                                     <char>
## 1: Pre (<6 months since LMP AND no prior bilateral ovariectomy AND not on estrogen replacement)
## 2:           Post (prior bilateral ovariectomy OR >12 mo since LMP with no prior hysterectomy)
## 3: Pre (<6 months since LMP AND no prior bilateral ovariectomy AND not on estrogen replacement)
## 4:           Post (prior bilateral ovariectomy OR >12 mo since LMP with no prior hysterectomy)
## 5:           Post (prior bilateral ovariectomy OR >12 mo since LMP with no prior hysterectomy)
##      tumor.stage ER.status PR.status HER2.status
##      <char>      <char>      <char>      <char>
## 1:  stage iia      <NA>      <NA>      <NA>
## 2:  stage iib Positive Positive Positive
## 3:  stage iia Positive Positive Negative
## 4:  stage iib Positive Positive Negative
## 5:  stage iia Positive Positive Positive
```

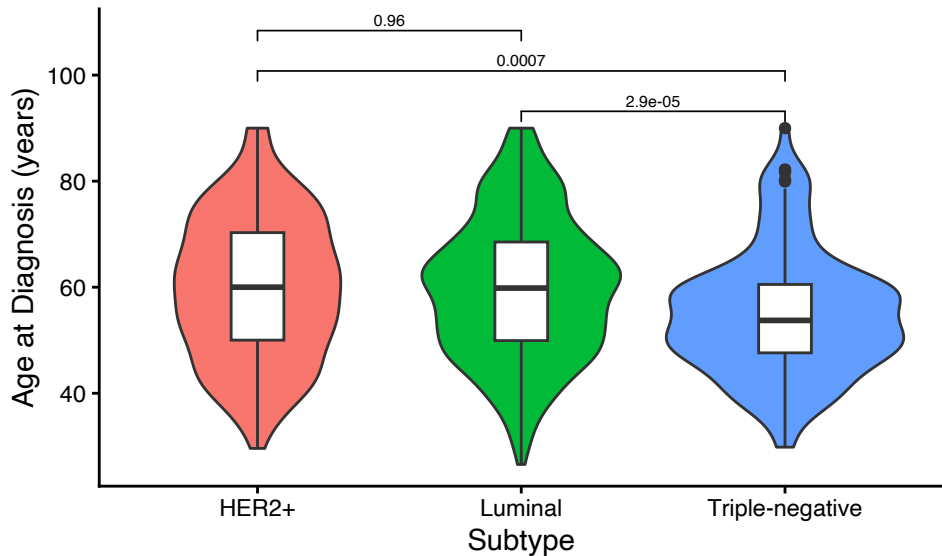
Age distribution by menopause status



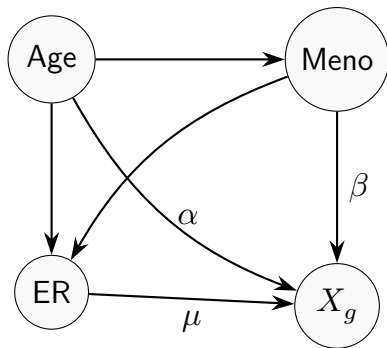
Age distribution by ER status



Age distribution by subtype



Probabilistic model



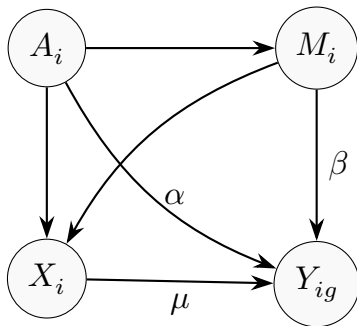
Observed variables:

- Age (continuous), Menopause (categorical), ER status (categorical)
- Gene expression matrix X_{gi} for a gene g in a sample i .

Parameters (open):

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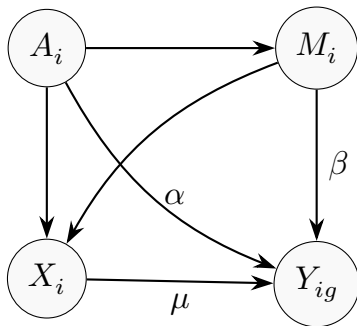
Let's focus on one sub-problem: ER+ vs. ER-



"divide and conquer"

- Instead of resolving all the parameters all at once, can we estimate each effect one by one?

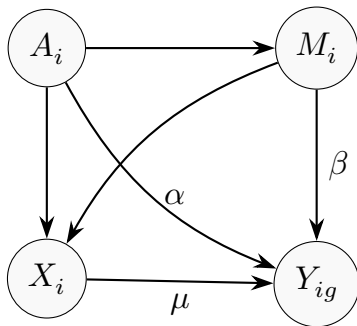
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"divide and conquer"

- Instead of resolving all the parameters all at once, can we estimate each effect one by one?
- Are there any genes **significantly changed** as a result of the ER status?

Let's focus on one sub-problem: ER+ vs. ER-



"divide and conquer"

- Instead of resolving all the parameters all at once, can we estimate each effect one by one?
- Are there any genes **significantly changed** as a result of the ER status?
- How can we **test** any significant changes?

Today's lecture

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- 2 Key concepts**
- 3 An example of hypothesis testing
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Recap: random variable (lect02)

Random Variable

- Mathematical representation of all possible values
- Denoted by uppercase letters: K , X , T
- Represents abstract set of possibilities

Observed Value

- Actual result from experiment
- Denoted by lowercase letters: k , x , t
- The specific measurement we obtained

Probability Distribution (lect02)

Definition: Map: possible value $\rightarrow$ probability

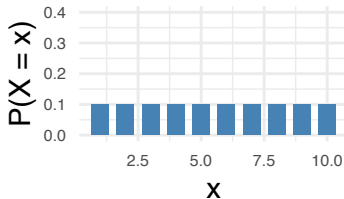
- Describes how frequently different values occur
- Discrete (finite) or continuous (infinite)

Examples:

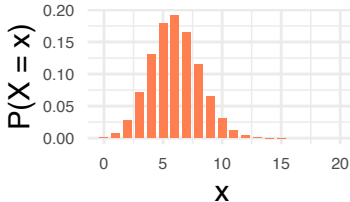
- Discrete: Binomial, Poisson
- Continuous: Normal (Gaussian) distribution

Common Discrete Distributions

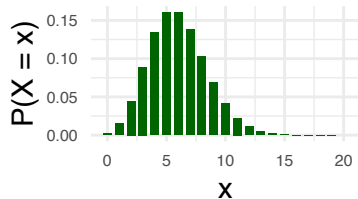
Discrete Uniform(1,10)



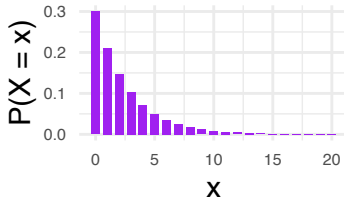
Binomial($n=20$, $p=0.3$)



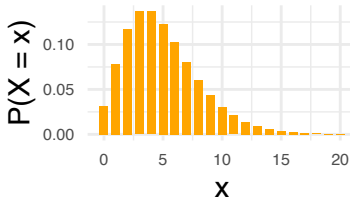
Poisson($\lambda=6$)



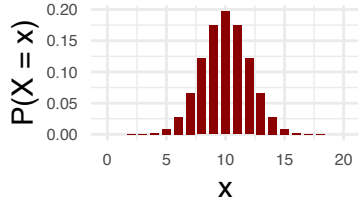
Geometric($p=0.3$)



Negative Binomial
($r=5$, $p=0.5$)

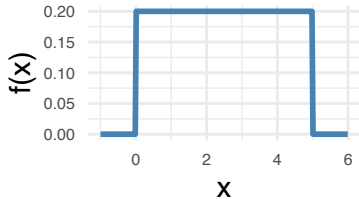


Hypergeometric
($m=50$, $n=50$, $k=20$)

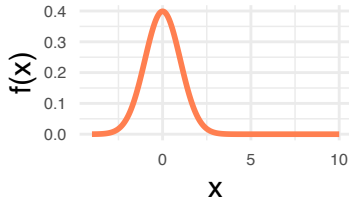


Common Continuous Distributions

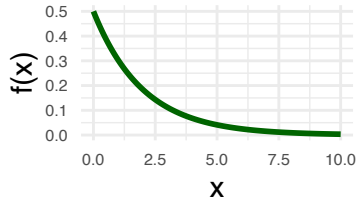
Continuous Uniform
($a=0$, $b=5$)



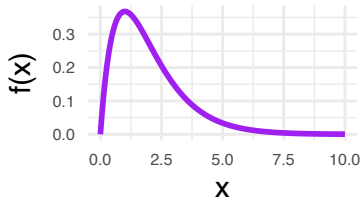
Normal($\mu=0$, $\sigma=1$)



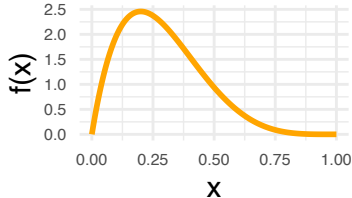
Exponential($\lambda=0.5$)



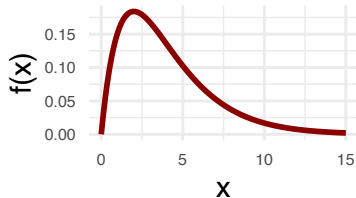
Gamma($\alpha=2$, $\beta=1$)



Beta($\alpha=2$, $\beta=5$)



Chi-squared($df=4$)



What is Hypothesis Testing?

A formal procedure for making decisions based on finite data samples

Key question: Is the observed effect real, or could it have occurred by chance?

Applications in genomics:

- Is a gene differentially expressed between conditions?
- Are two genetic variants associated with a trait?
- Does a drug treatment have an effect on gene expression?

First hypothesis test: The Lady Tasting Tea

- Dr. Muriel Bristol claimed she could tell whether milk or tea was poured first into a cup
- RA Fisher designed a randomized control experiment:
 - 8 cups of tea (4 milk-first, 4 tea-first)
 - Dr. Bristol tasted all 8 cups
 - She had to identify which 4 were milk-first

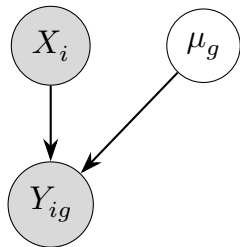


The question: Could she really tell the difference, or was she just guessing?

- H_0 : She's guessing (no ability)
- If guessing, probability of getting all 4 correct by chance = $1/\binom{8}{4} \approx 0.014$

https://en.wikipedia.org/wiki/Lady_tasting_tea

Let's focus on how X affects Y in the full PGM

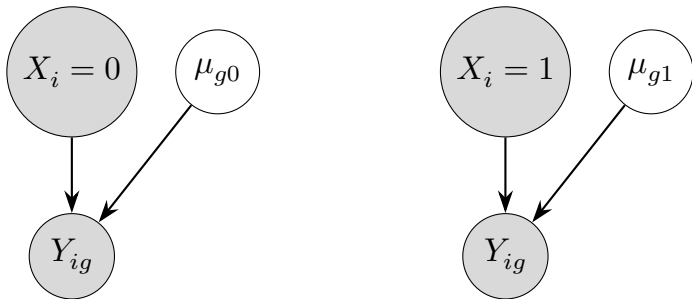


- X_i for each $i \in [n]$:

$$X_i = \begin{cases} 0 & \text{if ER+} \\ 1 & \text{if ER-} \end{cases}$$

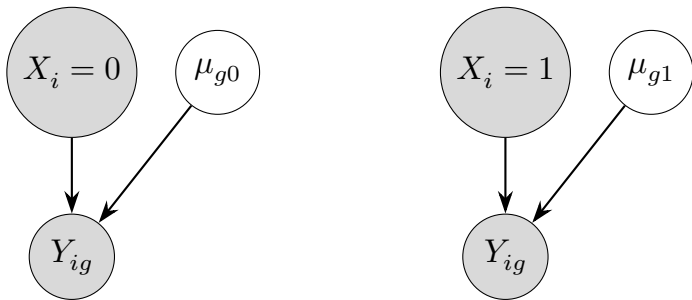
- We can say gene expression Y_{ig} changes as a function of X by a gene g 's μ_g
- What would be the probability distribution of Y ?

How significantly X affects Y ?



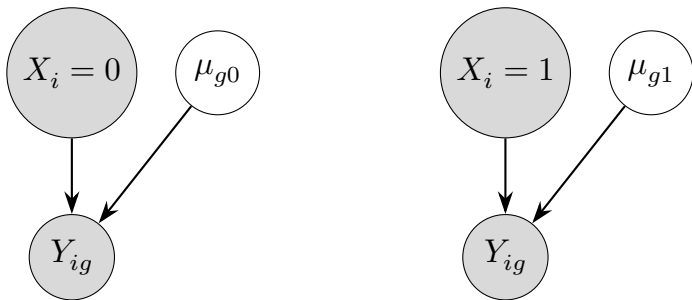
Here, we show the "data-generating" scheme more explicitly.

How significantly X affects Y ?



- 1 Has this gene g been affected by ER?
- 2 Y_g changes depending on $X = 0$ or $X = 1$?
- 3 Is $\mu_{g0} = \mu_{g1}$?

How significantly X affects Y ?



Formal writing of hypothesis testing:

$$H_0 : \mu_{g0} = \mu_{g1} \quad \text{vs.} \quad H_1 : \mu_{g0} \neq \mu_{g1}$$

Today's lecture

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Five Steps of Hypothesis Testing

① Choose Test statistic

Five Steps of Hypothesis Testing

- 1 Choose Test statistic
- 2 Establish Null Hypothesis (H_0)

Five Steps of Hypothesis Testing

- ① Choose Test statistic
- ② Establish Null Hypothesis (H_0)
- ③ Define "rejection" region

For Step 1-3: What would happen **under the null**?

Five Steps of Hypothesis Testing

① **Choose Test statistic**

② Establish Null

Hypothesis (H_0)

③ Define "rejection" region

4. Compute Test Statistic

5. Make a decision

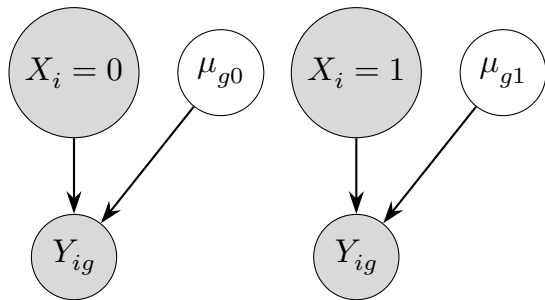
straightforward

For Step 1-3: What would happen **under the null**?

How to come up with test statistic?

- Decide on the effect of interest
- Design experiment or study
- **Modelling can guide us**
- Choose a data summary function and test statistic

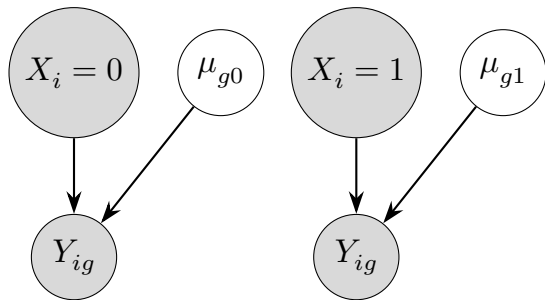
Choose test statistic: How significantly X affects Y ?



① What's our interest?

$$H_0 : \mu_{g1} = \mu_{g0} \quad \text{vs.} \quad H_1 : \mu_{g1} \neq \mu_{g0}$$

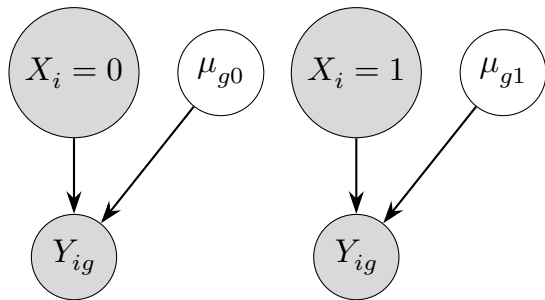
Choose test statistic: How significantly X affects Y ?



- 1 What's our interest?
- 2 The difference of the "average" parameters

$$H_0 : \mu_{g1} = \mu_{g0} \quad \text{vs.} \quad H_1 : \mu_{g1} \neq \mu_{g0}$$

Choose test statistic: How significantly X affects Y ?

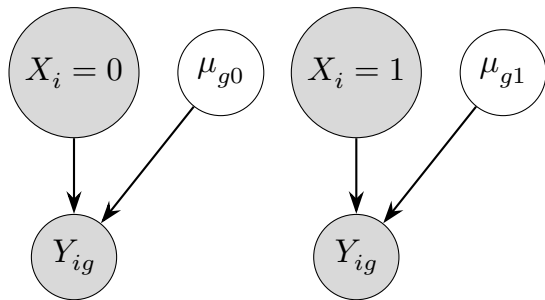


- 1 What's our interest?
- 2 The difference of the "average" parameters
- 3 How can we estimate the mean (1st moment) from data?

$$\{Y_{1g}, \dots, Y_{ng}\}$$

$$H_0 : \mu_{g1} = \mu_{g0} \quad \text{vs.} \quad H_1 : \mu_{g1} \neq \mu_{g0}$$

Step 1. Choose Test Statistics



$$H_0 : \mu_{g1} = \mu_{g0} \quad \text{vs.} \quad H_1 : \mu_{g1} \neq \mu_{g0}$$

- Sample mean for $X_i = 0$

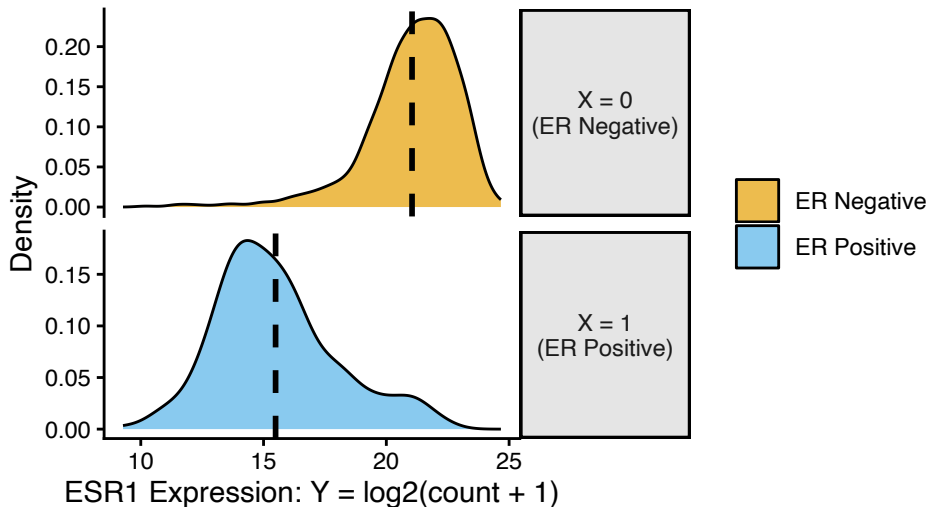
$$\bar{Y}_{g0} = \frac{\sum_{i=1}^n (1 - X_i) Y_i}{\sum_{i=1}^n (1 - X_i)}$$

- Sample mean for $X_i = 1$

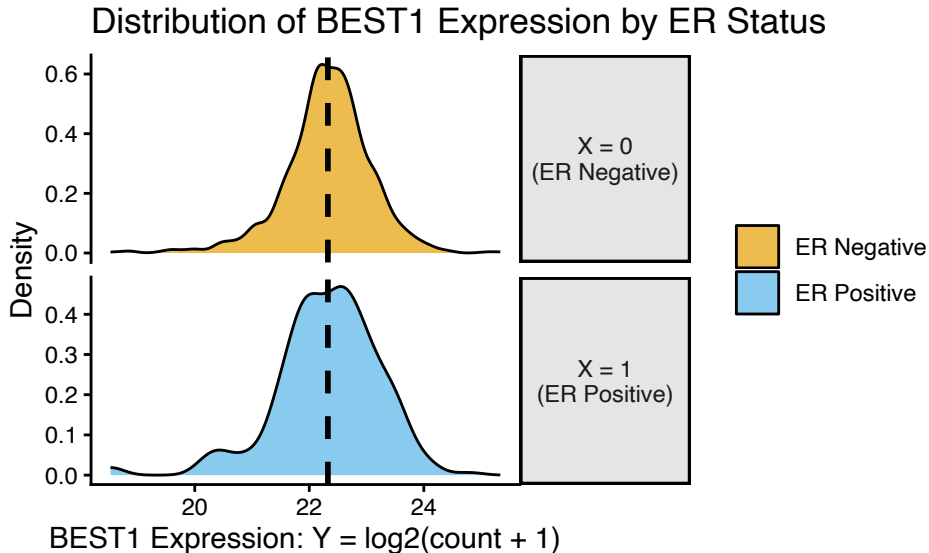
$$\bar{Y}_{g1} = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i}$$

Example: *ESR1* expression by ER status

Distribution of *ESR1* Expression by ER Status



Example: *BEST1* expression by ER status



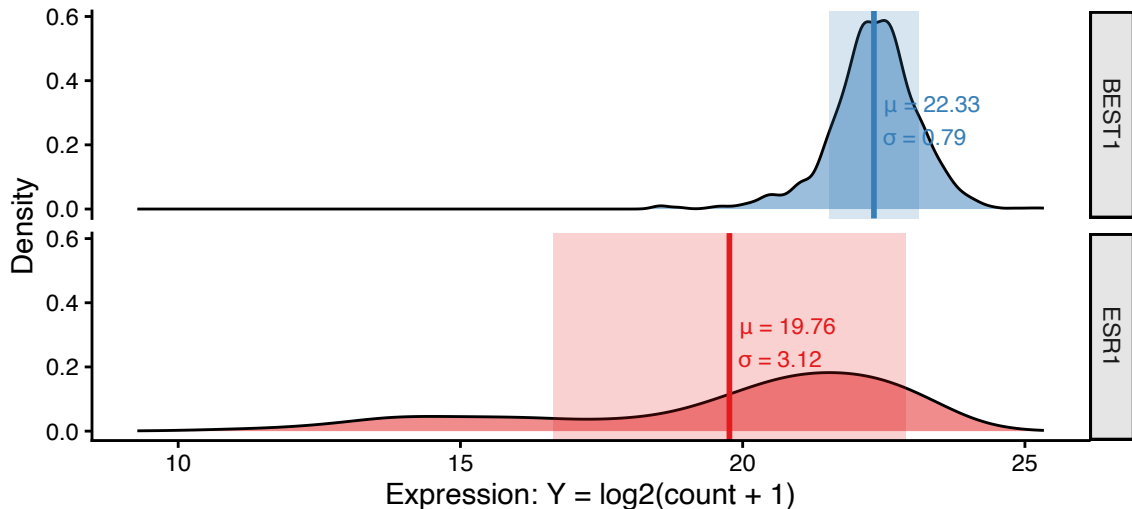
Define Test Statistics - idea 1

$$T = \bar{Y}_{g1} - \bar{Y}_{g0}$$

Maybe... or why not?

Compare variance: *ESR1* vs. *BEST1*

Variance Comparison: ESR1 vs BEST1 (Marginal Distributions)



Revised Test Statistic - should adjust T by variance

$$T = \frac{\bar{Y}_{g1} - \bar{Y}_{g0}}{\sqrt{\mathbb{V}[\bar{Y}_{g1} - \bar{Y}_{g0}]}}$$

Side note: Test statistic is a random variable

In this example,

$$T = \frac{\bar{Y}_{g1} - \bar{Y}_{g0}}{\sqrt{\mathbb{V}[\bar{Y}_{g1} - \bar{Y}_{g0}]}}$$

- Y is a random variable
- And the sample mean is a function of Y (also, random variable)
- In general, if X is a random variable, a function take on X , i.e., $g(X)$, is also a random variable.

Five Steps of Hypothesis Testing

- 1 Choose Test statistic
- 2 **Establish Null Hypothesis (H_0)**
- 3 Define "rejection" region
- 4 Compute Test Statistic
- 5 Make a decision

Step 2. Estimate/simulate null Distribution

Definition: The probability distribution of a test statistic when H_0 is true

- Represents expected outcomes if there were **no real effect**
- Used as baseline for comparison
- Allows us to assess how “extreme” our observed data is

Idea:

- If observed statistic is unlikely under null distribution, we have evidence against H_0
- Under the null hypothesis H_0 , what would be the distribution of test statistic T ?

How do we estimate the distribution of T ?

Option 1: Theoretical

- Make distributional assumptions
- Derive the null distribution mathematically
- Use known probability distributions

Example: t-distribution

- Assume normality
- Use Central Limit Theorem
- Analytical formula available

Pros: Fast, well-studied

Cons: Requires assumptions

Option 2: Empirical

- Simulate/resample data
- Generate null distribution empirically
- No distributional assumptions needed

Example: Permutation test

- Randomly shuffle group labels
- Recalculate test statistic
- Repeat many times

Pros: Assumption-free

Cons: Computationally intensive

Option 1: Theoretical approach (t-distribution)

For ESR1 gene:

```
# Perform t-test using theoretical t-distribution  
t.test(Y ~ X, data = .df_esr1, var.equal = TRUE)
```

How it works:

- 1 Assume $Y_{ig} \sim N(\mu_g, \sigma^2)$
- 2 Under H_0 : $T = \frac{\bar{Y}_1 - \bar{Y}_0}{s_{\text{pooled}} \sqrt{\frac{1}{n_0} + \frac{1}{n_1}}} \sim t_{n_0 + n_1 - 2}$
- 3 Look up p-value from t -distribution tables/functions

Option 2: Empirical approach (Permutation test)

For ESR1 gene:

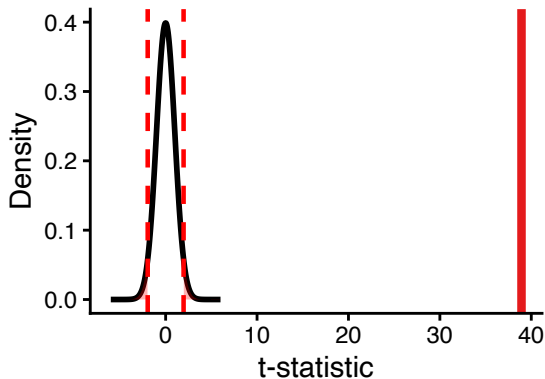
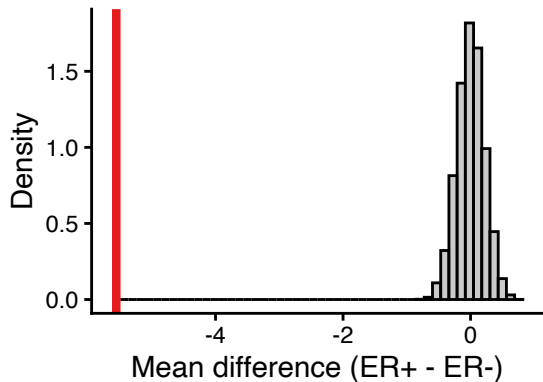
```
# Permutation test for ESR1  
set.seed(123)  
perm_result <- perm.test(.df_esr1, n_perm = 10000)
```

How it works:

- 1 Calculate observed difference: $T_{\text{obs}} = \bar{Y}_1 - \bar{Y}_0$
- 2 Shuffle group labels (X) many times
- 3 Recalculate T for each shuffle
- 4 P-value = proportion of shuffles with $|T| \geq |T_{\text{obs}}|$

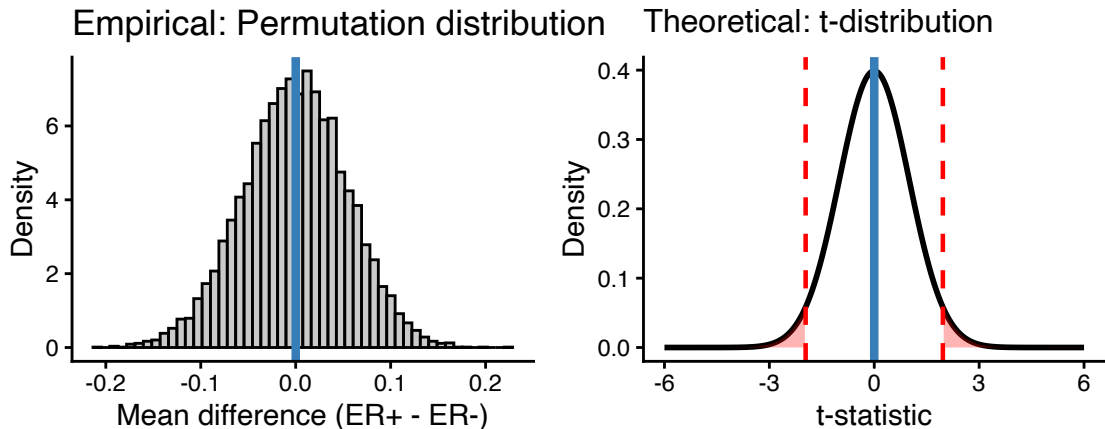
Comparing approaches: ESR1 (significant gene)

Empirical: Permutation distribution Theoretical: t-distribution



ESR1: Both approaches agree - highly significant ($p < 0.001$)

Comparing approaches: BEST1 (non-significant gene)



BEST1: Both approaches agree - not significant ($p > 0.05$)

When to use which approach?

Theoretical (t-distribution)

Use when:

- Sample size is moderate to large
- Data is approximately normal
- Speed is important
- Need confidence intervals

Advantages:

- Fast computation
- Well-established theory
- Standard in literature

Empirical (Permutation)

Use when:

- Sample size is small
- Data is non-normal
- Variance assumptions violated
- Custom test statistics

Advantages:

- No distribution assumptions
- Exact p-values (for small samples)
- Flexible

Five Steps of Hypothesis Testing

- 1 Choose Test statistic
- 2 Establish Null Hypothesis (H_0)
- 3 **Define "rejection" region**
- 4 **Compute Test Statistic**
- 5 Make a decision

Step 3. Define the rejection region

Definition: Set of test statistic values where we reject H_0

Determined by:

- **Significance level (α):** Acceptable false positive rate (typically 0.05)
- **Power considerations:** Maximize probability of detecting true effects
- **Alternative hypothesis type:**
 - Two-sided tests: Split rejection region between both tails
 - One-sided tests: Concentrate region on one tail

Choice reflects: What values provide “sufficient evidence” against H_0

Step 4. Two-sample t-statistic by sample mean & var.

- ① Calculate sample means for each group:

$$\bar{Y}_0 = \frac{1}{n_0} \sum_{i: X_i=0} Y_i \quad \text{and} \quad \bar{Y}_1 = \frac{1}{n_1} \sum_{i: X_i=1} Y_i$$

- ② Calculate sample variances for each group:

$$s_0^2 = \frac{1}{n_0 - 1} \sum_{i: X_i=0} (Y_i - \bar{Y}_0)^2 \quad \text{and} \quad s_1^2 = \frac{1}{n_1 - 1} \sum_{i: X_i=1} (Y_i - \bar{Y}_1)^2$$

Step 4. Two-sample t-statistic - continued

- ③ Calculate pooled variance (assuming equal variance):

$$s_{\text{pooled}}^2 = \frac{(n_0 - 1)s_0^2 + (n_1 - 1)s_1^2}{n_0 + n_1 - 2}$$

- ④ Compute the t-statistic:

$$T = \frac{\bar{Y}_1 - \bar{Y}_0}{s_{\text{pooled}} \sqrt{\frac{1}{n_0} + \frac{1}{n_1}}}$$

In practice, use a built-in `t.test` function

`t.test`

`package:stats`

R Documentation

Student's t-Test

Description:

Performs one and two sample t-tests on vectors of data.

Usage:

`t.test(x, ...)`

In practice, for many hypotheses, matrixTests

`ttest` package:matrixTests R Documentation

t-test

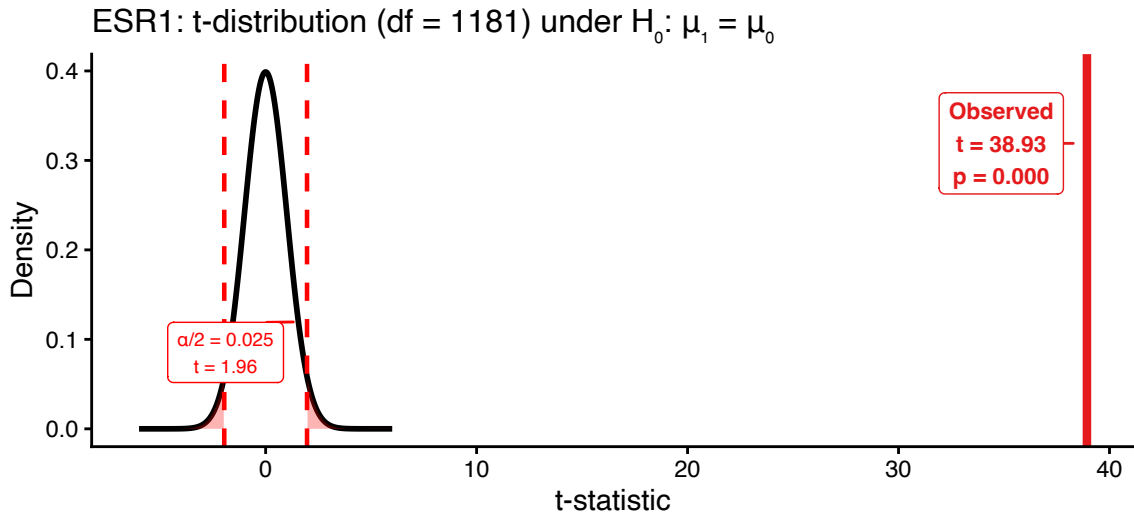
Description:

Performs a t-test on each row/column of the input matrix.

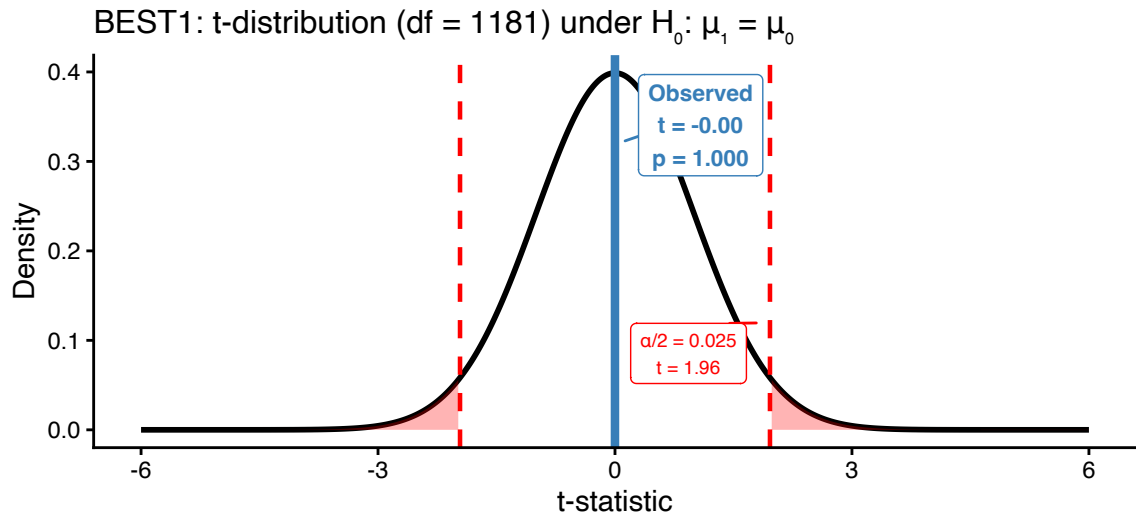
Usage:

```
row_t_equalvar(x, y, null = 0, alternative = "two.sided", conf.level = 0
```

t -distribution of *ESR1*



t -distribution of *BEST1*



P-value

Definition: Probability of observing a test statistic as extreme or more extreme than observed, assuming H_0 is true

- Scaled to range $[0, 1]$
- Smaller p-values = stronger evidence against H_0
- Common threshold: $p < 0.05$ for “statistical significance”

Important:

- p-value is NOT the probability that H_0 is true
- It's the probability of the data (or more extreme), given H_0

Five Steps of Hypothesis Testing

- 1 Choose Test statistic
- 2 Establish Null Hypothesis (H_0)
- 3 Define "rejection" region
- 4 Compute Test Statistic
- 5 **Make a decision**

Interpreting p-values

Which of the following are true? (select all that apply)

- ☒ a. If the effect size is very small, but the sample size is large enough, it is possible to have a statistically significant p-value
- ☒ b. A study may show a relatively large magnitude of association (effect size), but a statistically insignificant p-value if the sample size is small
- ☒ c. A very small p-value indicates there is a very small chance the finding is a false positive

Today's lecture

- 1 Motivation and Problem definition
- 2 Key concepts
- 3 An example of hypothesis testing
- 4 Discussion**

Types of Errors

	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct (Power)
Fail to reject	Correct	Type II Error (β)

Type I Error (False Positive)

- Reject H_0 when it's actually true
- Controlled by significance level α

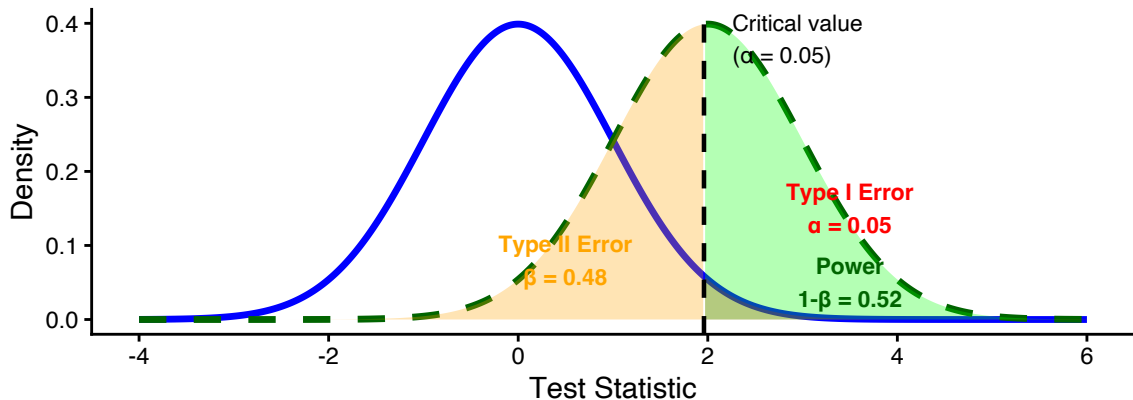
Type II Error (False Negative)

- Fail to reject H_0 when it's actually false
- Related to power: Power = $1 - \beta$

Visualizing Type I and Type II Errors

Type I Error, Type II Error, and Statistical Power

Distribution — $H_0: \mu = 0$ (Null) - - - $H_1: \mu = 2$ (Alternative)



Statistical Power

Definition: Probability of correctly rejecting a false H_0

- Measures ability to detect real effects when they exist
- Power = $1 - \beta$, where β is Type II error rate

Factors affecting power:

- Sample size (larger $\rightarrow$ higher power)
- Effect size (larger $\rightarrow$ higher power)
- Significance level α (larger $\rightarrow$ higher power)
- Variability in data (lower $\rightarrow$ higher power)

Common p-value pitfalls

A p-value is **NOT**:

- The probability that the null hypothesis is true
- The probability that the finding is a “fluke”
- A measure of the size or importance of observed effects

Different kinds of t -tests:

- One sample *or* **two samples**
- One-sided *or* **two sided**
- Paired *or* **unpaired**
- Equal variance *or* unequal variance

Common Non-parametric Tests

Wilcoxon Rank-Sum Test (Mann-Whitney U)

- Non-parametric alternative to two-sample t-test
- Tests if two distributions differ

Wilcoxon Signed-Rank Test

- Non-parametric alternative to paired t-test

Permutation Tests

- Simulate null distribution

Final quiz: True or False

- Modelling doesn't help us design hypothesis testing
- We need complete knowledge to come up with a working hypothesis
- p-value is the evidence for the null hypothesis
- A small p-value suggests our alternative hypothesis H_1 is true
- It is highly recommended to implement my own hypothesis tests
- Hypothesis testing is an ultimate scientific proof

Some aspect of scientific explanation

"Explaining a contrast is sometimes easier than explaining the fact"

"P rather than Q" vs. "P" or "Q" alone

(Lipton 1991)

Multiple lines of orthogonal evidence

How can we overcome the limitation of hypothesis testing?



(Munafò and Davey Smith 2018)

Yongjin Park (slide credit: Keegan Korthauer)

Hypothesis Testing

14 January 2026

Summary

- Modelling $\implies$ expose a specific hypothesis
- We may **NOT** have complete knowledge before hypothesis
- Testing **ONLY** reject the null hypothesis
- Use existing (R) hypothesis test functions whenever possible

References

- Cancer Genome Atlas Network. 2012. “Comprehensive Molecular Portraits of Human Breast Tumours.” *Nature* 490 (7418): 61–70.
- EBCTCG, C Davies, J Godwin, et al. 2011. “Relevance of Breast Cancer Hormone Receptors and Other Factors to the Efficacy of Adjuvant Tamoxifen: Patient-Level Meta-Analysis of Randomised Trials.” *Lancet* 378 (9793): 771–84.
- Lipton, Peter. 1991. “Contrastive Explanation and Causal Triangulation.” *Philos. Sci.* 58 (4): 687–97.
- Munafò, Marcus R, and George Davey Smith. 2018. “Robust Research Needs Many Lines of Evidence.” *Nature* 553 (7689): 399–401.
- Osako, Tomo, Hakwoo Lee, Gulisa Turashvili, et al. 2020. “Age-Related Protein and Transcript Expression in Breast Cancer and Normal Breast Tissues Is Dominated by Host Endocrine Effects.” *Nature Cancer* 1 (5): 518–32.